



Trium Idea Intelligence Platform

Product Requirements Document

Benchmarking | Local viability | Daily idea and policy scouting

Version	0.1 (Draft for review)
Date	30 September 2026
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Status	Working name; scope confirmed for Flows 1, 2a, 2b, 2c

1. Summary

Trium needs one platform that does two jobs. First, it benchmarks an idea against what has been built globally, with deliberate coverage of emerging and nearby markets, and tells the team what to apply and what to avoid in Nigeria. Second, it screens ideas for Nigerian viability, compares them with the Vanta idea bank, pre-grades the survivors against Trium's existing criteria, and delivers them either on demand or automatically every day.

Four flows are in scope:

Flow	What it does	Trigger	Output
1 Global benchmarking	Researches an idea across global, emerging and nearby markets and extracts comparable benchmarks	On demand (prompt or upload)	Report in PDF, Word or PPT (user chooses) with an Apply / Avoid summary for Nigeria
2a Local viability (on demand)	Extracts ideas from a prompt or document, removes ones already in Vanta, tests the rest for Nigerian viability, drafts and grades submission responses	On demand	Graded, submission-ready entries
2b Emerging-market scout	Scrapes 30 to 50 emerging-market tech sources daily (Nigeria data excluded), then runs the 2a pipeline	Scheduled daily	Email to DIT for ideas graded B to A
2c Nigerian policy scout	Scrapes Nigerian news, government and legal sources for new and incoming policy, derives ideas, then runs the 2a pipeline	Scheduled daily	Email to DIT for ideas graded B to A

Key decisions made in this document

- **One shared pipeline.** Flows 2a, 2b and 2c differ only in how ideas enter. Extraction, dedupe, viability, drafting, grading and delivery are built once.
- **2b source selection: hybrid.** A curated, human-approved source registry runs every day. A monthly AI-assisted discovery job proposes new sources for approval. Reasons are in section 6.3.
- **Free tools only.** Gemini free API tier as the primary LLM, open-source scraping and parsing libraries as the core, and free search tiers (Tavily, Exa) for discovery. The full verdict on every tool you named is in section 8.
- **Flow 1 is built first.** It has no Vanta dependency, so work can start while Vanta access is being arranged.
- **Vanta is read-only in v1.** The platform reads the idea bank and grading data. Writing entries back is a later phase.

2. Problem, goals and non-goals

Benchmarking today is manual and uneven. Coverage of emerging and nearby markets depends on who does the research. Ideas from other markets are spotted by chance, and there is no systematic check that an idea is new to the Vanta idea bank, viable in Nigeria, and likely to pass Trium's assessment before anyone spends time on it.

Goals

- Produce a sourced benchmark report for any idea in minutes rather than days, with emerging and nearby markets always represented where evidence exists.
- Screen global ideas for Nigerian viability and filter out duplicates of Vanta entries without discarding near-similar ones that differ in a meaningful way.
- Deliver a daily flow of pre-assessed, submission-ready ideas from emerging markets and from Nigerian policy change.
- Run at zero software cost at launch.

Non-goals

- Replacing human assessors. The AI pre-screens and drafts; people decide.
- Writing entries into Vanta automatically in v1.
- Scraping paywalled, login-gated or personal-data sources.

- Financial modelling of the ideas. Benchmarks report only what is publicly disclosed.

3. Users

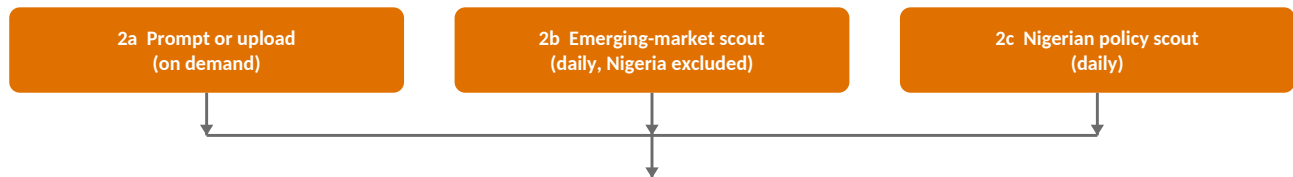
User	Needs
Venture and innovation team (primary)	Run benchmarks and screen documents on demand; choose the output format; review the daily shortlist
DIT (email recipient)	Receive only strong ideas (B to A) in a consistent, structured email with enough evidence to act on
Assessors	Trust that AI grading follows the Vanta criteria and their own past patterns; see where AI and human grades differ
Platform admin	Manage the source registry, thresholds, schedules and quota health

4. Platform overview

FLOW 1 Global benchmarking (standalone)



FLows 2a / 2b / 2c Entry points (different front end)



SHARED IDEA PIPELINE (build once, used by all three)



Vanta (read-only in v1): idea bank, submission fields, grading criteria, past assessments

Every output carries source URLs, retrieval dates and a confidence flag.

Figure 1. Flow 1 is standalone. Flows 2a, 2b and 2c share one pipeline and differ only at the entry point.

Working principles (apply to every flow)

- Source-grounded.** Every figure, claim and benchmark carries a source URL and retrieval date. If a number cannot be found, the output says so. No estimated or inferred figures are presented as fact.
- Nigerian grounding.** Viability and Apply / Avoid guidance must reference the Nigerian context: regulators, payment rails, infrastructure, FX and affordability.
- Human in the loop.** Every AI judgement (duplicate, viability, grade) is shown with its reasoning and can be overridden.
- Honest drafting.** Drafted submission responses must improve clarity and structure using sourced facts. They must not inflate claims to reach a score; assumptions are labelled as assumptions.

5. Flow 1: Global benchmarking

5.1 Steps

#	Step	Requirement
1	Receive input	FR-1.1 Accept a typed prompt, or an uploaded PDF, DOCX, PPTX or TXT. FR-1.2 Extract text locally (pdfplumber, python-docx, python-pptx); use Tesseract OCR for scans; use the LLM only to structure the text.
2	Understand the idea	FR-1.3 Produce a one-paragraph idea brief (problem, solution, customer, business model, sector) for the user to confirm or edit before research starts.
3	Scan online sources	FR-1.4 Generate region-specific search queries and run them through Tavily and Exa, then fetch and parse the pages. FR-1.5 Cover three tiers: global leaders, emerging markets, and nearby markets. FR-1.6 Where a tier returns nothing, state that explicitly rather than padding.
4	Extract benchmarks	FR-1.7 Fill the benchmark record (table below) for each company or initiative found, with source and confidence per field. FR-1.8 Mark fields as 'not disclosed' instead of guessing.
5	Synthesise	FR-1.9 Comparison table across benchmarks. FR-1.10 Patterns: what worked, what failed, recurring business models. FR-1.11 A dedicated Apply in Nigeria and Avoid in Nigeria section tied to Nigerian regulators, infrastructure and market conditions.
6	Output	FR-1.12 User picks PDF, Word or PowerPoint at request time. FR-1.13 Use the Trium template (orange, charcoal, Lato, wordmark). FR-1.14 Include a sources appendix.

5.2 Benchmark record

Group	Fields
Identity	Company or initiative, country, region tier (global / emerging / nearby), launch year, status (active, pivoted, shut down)
Numbers	Funding raised, users or customers, revenue or GMV, growth, unit economics. Only as publicly disclosed, with date
Coverage	Geographies served, segments, channels and distribution partners
Business model	Revenue streams, pricing, regulatory route or licence, key partnerships
Lessons	What worked, what did not, reasons for pivot or failure
Evidence	Source URLs, retrieval date, confidence (High / Medium / Low), conflicting sources if any

5.3 Coverage rules

- **Emerging markets** (working list): India, Indonesia, Philippines, Vietnam, Pakistan, Bangladesh, Brazil, Mexico, Colombia, Egypt, Turkey.
- **Nearby markets** (working definition): Ghana, Kenya, South Africa, Egypt, Cote d'Ivoire, Senegal, Rwanda, Ethiopia, Tanzania, Uganda. To be confirmed by Trium.
- Target at least three benchmarks per tier where they exist, configurable per request. Non-English sources (French, Portuguese, Spanish) are read and translated by the LLM.

6. Flows 2a, 2b and 2c

6.1 Shared idea pipeline

This is the back half used by all three flows. Each stage stores its result and reasoning so that any idea can be traced from source to final grade.

Stage	Requirement
Extract ideas	FR-2.1 Identify each distinct idea in the input and normalise it into the idea schema (Appendix A). FR-2.2 A document may contain several ideas; each is processed separately.

Stage	Requirement
Dedupe vs Vanta	FR-2.3 Read all Vanta idea-bank entries. FR-2.4 Shortlist likely matches by embedding similarity, computed locally so idea-bank content is not sent to an external API. FR-2.5 An LLM judge compares each candidate pair and returns one of three verdicts: Exact duplicate (drop, link to the Vanta entry), Near-similar (keep; record the differentiator and the Vanta entry it resembles), New (keep). FR-2.6 Only exact duplicates are dropped. When unsure, the verdict is Near-similar.
Nigeria viability	FR-2.7 Score each new or near-similar idea Low / Medium / High using the rubric below, with a written rationale and sources. FR-2.8 Keep Medium or higher (threshold configurable).
Draft responses	FR-2.9 For each survivor, draft entry-submission responses in the Vanta field structure, using only sourced facts and labelled assumptions.
Grade	FR-2.10 Grade the draft against the live Vanta criteria. FR-2.11 Retrieve the most similar previously assessed ideas, with their human scores and comments, as calibration examples. FR-2.12 If below pass, allow up to two revision rounds; if still below pass, park the idea with the gaps listed. Do not email or promote it.
Deliver	2a: on-screen results and export. 2b and 2c: automatic email (section 6.4).

Viability rubric (working version)

Dimension	Question asked about the Nigerian market
Demand and affordability	Is there a paying customer, and does the price work at local incomes?
Regulation	Which regulators apply (for example CBN, SEC, NCC, NDPC, FCCPC) and is there a licensing path?
Infrastructure and payments	Does it depend on connectivity, power, identity or payment rails that are reliable here?
Competition	Who already serves this, and is there room for a differentiated entrant?
Unit economics and FX	Do margins survive naira pricing, FX exposure and local cost of delivery?
Distribution and trust	Can the product reach users through channels and partners that exist locally?

6.2 Flow 2a: on-demand viability screening

- **FR-2a.1** Accept a prompt or an uploaded document, parsed as in Flow 1.
- **FR-2a.2** Run the shared pipeline and show, for every extracted idea, its fate: dropped as exact duplicate (with the matching Vanta entry), parked below viability, parked below pass, or passed.
- **FR-2a.3** Output results graded on the Vanta scale, exportable as PDF, Word or PPT.

6.3 Flow 2b: emerging-market scout (daily)

Recommendation: hybrid source management. Keep a curated source registry that the daily job reads, and add a monthly discovery job that proposes new sources for human approval. This is better than deciding the top sites afresh each day because it is deterministic (the same sources are scanned, so day-to-day changes are real changes), it uses free RSS feeds and sitemaps instead of burning limited search credits, and it is auditable when someone asks why an idea was or was not found.

- **FR-2b.1 Registry.** Each source has region, URL, feed or sitemap URL, tier, and health status. Tier A is emerging-market tech coverage (target 30 to 50). Tier B is a ranked pool of global sources (for example Crunchbase News, TechCrunch, Sifted).
- **FR-2b.2 Fallback rule.** Each day, count healthy Tier A sources. If below the target, fill the gap from Tier B in rank order, skipping any source already scanned that day.
- **FR-2b.3 Daily scan.** Read new items from feeds and sitemaps, extract article text, and discard anything already processed (by URL and content hash).
- **FR-2b.4 Nigeria exclusion.** Tag each item's markets. Drop items whose subject is Nigeria. For multi-country items, keep only the non-Nigerian information.
- **FR-2b.5 Ideas.** Extract ideas into the idea schema, then run the shared pipeline.

- **FR-2b.6 Trigger and action.** When an idea reaches the email threshold (default grade B or above), send a structured email to the DIT mailbox automatically. Delivery by Resend, which Vanta already uses.
- **FR-2b.7 Monthly discovery.** Use Exa and Tavily to find candidate sources; an admin approves before they enter the registry.
- **FR-2b.8 Health.** Flag sources that fail three runs in a row or return no new items for 30 days.

Structured email (per idea)

Section	Content
Subject	Idea name, grade, source market
Idea	Problem, solution, target customer, business model (Trium structure)
Origin	Where it was found, source link, date
Nigeria viability	Rating, rationale, key regulatory and infrastructure notes
Vanta grade	Overall grade and per-criterion scores with reasons
Draft submission	Drafted responses, ready to paste into Vanta
Similar entries	Any Near-similar Vanta entries and the stated differentiator
Action	Link to review or reject in the platform

Option: a single daily digest instead of one email per idea, selectable in settings.

6.4 Flow 2c: Nigerian policy scout (daily)

Flow 2c is 2b with a different source set and one extra step at the front: policy is turned into an opportunity before it becomes an idea.

- **FR-2c.1 Sources.** Registry categories: Nigerian news and business media, government and regulator websites (including the Official Gazette), and law firm and legal-update sites. Candidate lists are validated in Phase 0.
- **FR-2c.2 Policy record.** For each item extract: policy or rule name, issuing body, status (proposed, enacted, effective date), affected sectors, obligations created, and source link. Distinguish incoming policy from policy already in force.
- **FR-2c.3 Opportunity hypothesis.** For each policy, generate candidate ideas that the change creates (for example a compliance tool, a new market or a new distribution route) and cite the clause or statement each one relies on.
- **FR-2c.4** From here the flow is identical to 2b: schema, dedupe, viability, draft, grade, email at B to A.
- **FR-2c.5** The email adds a Policy section: policy name, body, status and link.

7. Vanta integration

Vanta is the source of truth for the idea bank, the submission fields and the grading criteria. The platform must read them live rather than keep its own copy of the rules.

Need	Requirement
Idea bank	FR-V.1 Read all idea-bank and portfolio entries, including name, description, problem, solution and stage, for dedupe.
Field structure	FR-V.2 Discover the submission fields from Vanta so the idea schema matches exactly. Appendix A is a working schema until this is done.
Grading criteria	FR-V.3 Load the current criteria, weights and grade scale from Vanta. Do not hard-code them.
Past assessments	FR-V.4 Read historical human assessments (scores and comments) for calibration examples.
Access method	FR-V.5 Preferred: a read-only API or service account. Fallback: a scheduled JSON or CSV export. Credentials held as secrets, never in code.
Write-back	FR-V.6 Out of scope for v1. Phase 5 can create draft entries in Vanta after human approval.

Need	Requirement
AI assessment reuse	FR-V.7 If Vanta's AI-assisted assessment is callable, use it for grading. If not, re-implement it from the same criteria; the TriumAssess rubric and prompt logic are the starting point, swapped to the free LLM.

8. Free tool selection

Free-tier limits change often and were not verified live for this draft. Phase 0 includes confirming every quota and data-use term before build. Only tools with a genuine free option are selected.

8.1 Language model

Tool	Role	Verdict
Gemini API (Google AI Studio free tier)	Primary LLM: extraction, viability, drafting, grading. Reads PDFs natively. Large context for long articles and documents	Selected
Groq and OpenRouter free models	Fallback and bulk classification (for example Nigeria tagging of articles) when the primary quota is reached	Selected (fallback)
Local embeddings (open-source sentence-transformers)	Dedupe similarity against Vanta without sending idea-bank content out	Selected

8.2 Search, scraping and automation

Tool	Fit for this platform	Verdict
Tavily	Search API built for LLMs. Good for Flow 1 discovery and monthly source discovery. Monthly free credit allowance	Selected
Exa	Semantic search and 'find similar' companies. Strong for benchmark discovery. Limited free monthly requests	Selected
Open-source core: feedparser, trafilatura, Playwright, Crawl4AI	Reads RSS and sitemaps, extracts clean article text, handles JavaScript pages. No credit limits	Selected (daily scraping backbone)
Firecrawl	Clean markdown extraction. Hosted free credits are limited; the open-source version can be self-hosted	Selected (reserve) for hard pages only
Apify	Ready-made scrapers. Small recurring free credit	Reserve for sites with no feed
Browse AI	No-code robots. Free credits too low for 30 to 50 sites daily	Not selected
Bardeen	Browser-side automation, not built for server-scheduled scraping	Not selected
Clay	Sales lead enrichment, not article or policy scraping	Not selected

8.3 Turning documents into data

Tool	Fit	Verdict
pdfplumber, pypdf, python-docx, python-pptx, Tesseract OCR	Free and unlimited. Extract text and tables locally, OCR scans	Selected
Gemini document understanding	Structures messy text into the idea schema	Selected
Mindee	Free tier is small and aimed at invoices and receipts	Optional fallback only
Airparser, Parseur, Docparser	Template and email parsers; free tiers are tiny or trial-only	Not selected
Google Document AI	Pay-as-you-go with trial credit, not a lasting free tier	Not selected (fails free-only rule)

8.4 Supporting stack (all free)

Layer	Choice
Backend	Python with FastAPI. The scraping, parsing and report-generation libraries are all Python
Scheduler	GitHub Actions cron for the two daily jobs. Free minutes cover a feed-based run of 30 to 50 sources
Database	Postgres with pgvector (for example Supabase free tier), or SQLite to begin
Email	Resend, already in use for Vanta
Report generation	reportlab or WeasyPrint (PDF), python-docx (Word), python-pptx (PowerPoint)
Front end	Simple React or Next.js app on Vercel or Netlify. Check that the free plan's terms allow company use

9. Non-functional requirements

Area	Requirement
Accuracy	100% of figures carry a source and date. Unknown values shown as 'not disclosed'. Low-confidence items visibly flagged
Free-tier resilience	Request queue with back-off, caching of fetched pages, per-provider quota dashboard, automatic fallback to the secondary LLM, and graceful partial runs when a quota is hit
Reliability	Daily jobs are idempotent and can be re-run safely. A failed source never fails the whole run
Performance (targets)	Flow 1 report within about 10 minutes. Flow 2a document within about 5 minutes. Daily jobs finish before start of business in Lagos
Confidentiality	Free LLM tiers may use submitted data to improve their products. Unpublished Trium content must not be sent until the provider's terms are confirmed. Dedupe embeddings run locally and only the closest candidate pairs go to the LLM judge
Legal and ethical	Respect robots.txt and site terms. Public pages only. Store summaries and links, not republished articles. No personal data collection
Auditability	Store prompts, model used, sources and reasoning for every output
Access	Login restricted to Trium accounts; roles for user and admin
Languages	Read English, French, Portuguese and Spanish sources

10. Risks and mitigations

Risk	Impact	Mitigation
Free-tier quotas change or run out	Daily jobs stall	Feed-first scraping, caching, fallback LLM, quota alerts, and a Phase 0 check of current limits
Hallucinated figures in benchmarks	Loss of trust	Source-grounded extraction, 'not disclosed' default, and citation check before output
Wrong duplicate call drops a good idea	Missed opportunity	Only exact duplicates dropped; uncertainty resolves to Near-similar; dropped items logged and reviewable
AI grades drift from human assessors	Bad shortlist	Calibration with past assessments; track AI vs human difference; admin review sample each month
Vanta access is delayed or restricted	2a, 2b, 2c blocked	Build Flow 1 first; accept a scheduled export as a fallback
Sites block scrapers or change layout	Coverage gaps	Feeds and sitemaps first, source health alerts, Apify reserve
Idea-bank data exposure to free LLM vendors	Confidentiality breach	Local embeddings, minimal excerpts to the judge, written approval of provider terms
Gaming the pass score	Inflated drafts	Honest-drafting rule and human review before anything enters Vanta

11. Success metrics (proposed targets)

Metric	Target
Benchmark reports with at least one emerging and one nearby market record (where evidence exists)	90% or higher
Claims with a source and date	100%
Exact-duplicate false positives on a 50-pair test set	Near zero; reviewed by the team
AI grade within one grade step of human grade on back-test	80% or higher
Daily job success rate	95% or higher
Emailed ideas DIT considers worth pursuing after 60 days	50% or higher

12. Delivery plan (indicative)

Phase	Scope	Weeks
0 Discovery	Vanta schema, access and grading criteria; past assessment export; validate source lists; confirm free quotas and data terms; build a 50-pair dedupe test set	1 to 2
1 Flow 1	Ingestion, research, benchmark extraction, Apply / Avoid summary, PDF, Word and PPT output	3 to 6
2 Shared pipeline and 2a	Vanta connector, dedupe, viability, drafting, grading, on-demand screening	7 to 10
3 Flow 2b	Source registry, daily scan, Nigeria exclusion, fallback rule, email trigger	11 to 14
4 Flow 2c	Policy sources, policy records, opportunity hypotheses, policy email section	15 to 17
5 Hardening	Vanta write-back (draft entries), digest option, feedback loop from assessors, monthly source discovery	18 onward

13. Open questions

- **Vanta access:** what read access is possible (API, database role or export), and who approves it?
- **Grading:** what is the exact Vanta scale and the pass mark? This document assumes letter grades with B to A as the email threshold.
- **DIT:** confirm the recipient address or list, and per-idea email versus daily digest.
- **Nearby markets:** confirm the list in section 5.3.
- **Data terms:** is Trium comfortable sending non-public idea content to a free LLM tier, or should sensitive steps use a locally run open model?
- **Source approval:** who signs off the source registry and monthly additions?
- **Incoming policy:** does 'incoming' include proposed bills and regulator exposure drafts, or only announced rules?

Appendix A. Working idea schema

Replace with the actual Vanta submission fields in Phase 0.

Field	Description
Idea name	Short, distinctive name
Problem	Who has the problem, how severe, how it is handled today
Solution	What is offered and how it works
Target customer	Segment and size

Field	Description
Business model	Revenue streams and pricing
Sector	Sector and sub-sector
Origin	Source market, company or policy, date found, link
Nigeria viability	Rating, rationale, sources
Vanta grade	Overall and per-criterion, with reasoning
Status	New, Near-similar (with Vanta entry), Exact duplicate, Parked, Passed